

Arpit Kumar

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Education

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| Indian Institute of Information Technology, Allahabad
<i>Master of Technology in Information Technology (CGPA: 8.24 / 10.00)</i> <ul style="list-style-type: none">• Relevant Coursework: Cryptography, Cloud & Edge Computing, Social Network Analysis, IoT | August 2023 - June 2025
Prayagraj, UP |
| Bhagalpur College of Engineering, Bhagalpur
<i>Bachelor of Technology in Computer Science & Engineering (CGPA: 8.18 / 10.00)</i> <ul style="list-style-type: none">• Relevant Coursework: Data Structures, Operating Systems, DBMS, Computer Networks, OOPs | August 2018 - June 2022
Bhagalpur, Bihar |

Projects

- Real-Time Crowd Behavior Analysis and Anomaly Detection** | *Deep Learning, Machine Learning*
- Developed a model which predicts anomalies like fighting, valence, abnormal behavior, and patterns in crowded places.
 - Implemented using algorithm CAE, leveraging mean square error loss function and Adam optimizer.
 - Tested the model on the UCSD dataset with 34 training videos and 36 testing videos, with each video divided into 200 frames.
 - Calculated regularity scores for evaluation: scores decrease for abnormal patterns and increase for normal patterns.
- Voice Assistant** | *Python, NLP*
- Developed a voice assistant in Python that listens to voice inputs, understands queries, and responds with appropriate answers.
 - Integrated built-in APIs to handle complex questions, convert text to speech, and perform real-time web automation.
 - Incorporated Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) techniques for enhanced understanding and response generation.

Skills

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| Languages: C, C++, Python | Deep Learning: CAE, LSTM, ResNet, Transformers |
| Tools: Visual Studio Code, Google Colab | Machine Learning: Basics of ML, DL, NLP, and LLMs |
| Database: SQL | Frameworks: Hugging Face, LangChain |

Experience

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| Indian Institute of Information Technology, Allahabad
<i>Teaching Assistant</i> <ul style="list-style-type: none">• Assisted in teaching Database Management Systems (DBMS) by conducting tutorials, preparing assignments, and explaining core concepts like normalization, indexing, and query optimization.• Supported Operating Systems (OS) course by helping students understand topics such as process synchronization, memory management, and scheduling algorithms through problem-solving sessions.• Evaluated assignments and exams, and guided students in course-related projects to enhance their analytical and practical skills. | August 2023 - June 2025 |
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Achievements and Certificates

- Qualified **GATE 2023** with a score of 468
- Solved over **600 questions** across various platforms like LeetCode, GeeksforGeeks (GFG) & Coding-Ninjas
- **CISCO Certifications:** Programming Essentials in Python & Networking Essentials
- **Publications:** Published a Paper at ICCCNT 2025, held at IIT Indore.